



The Surface Mobility of Glasses

Fei Chen *et al.*

Science **343**, 975 (2014);

DOI: 10.1126/science.1248113

This copy is for your personal, non-commercial use only.

If you wish to distribute this article to others, you can order high-quality copies for your colleagues, clients, or customers by [clicking here](#).

Permission to republish or repurpose articles or portions of articles can be obtained by following the guidelines [here](#).

The following resources related to this article are available online at www.sciencemag.org (this information is current as of February 27, 2014):

Updated information and services, including high-resolution figures, can be found in the online version of this article at:

<http://www.sciencemag.org/content/343/6174/975.full.html>

A list of selected additional articles on the Science Web sites **related to this article** can be found at:

<http://www.sciencemag.org/content/343/6174/975.full.html#related>

This article **cites 13 articles**, 3 of which can be accessed free:

<http://www.sciencemag.org/content/343/6174/975.full.html#ref-list-1>

This article appears in the following **subject collections**:

Materials Science

http://www.sciencemag.org/cgi/collection/mat_sci

pete fire ants (8). A fire ant, when drenched with formic acid, has no apparent defense and falls over dead (9). In contrast, the crazy ant, when dabbed with fire ant venom, will use its own weapon as its ultimate defense: It rinses itself clean with its own formic acid. LeBrun *et al.* show that 98% of crazy ants survive fire ant venom; when they are experimentally denied access to their formic acid, survival drops to 48%. Moreover, when crazy ants competed against eight Texas ant species, each of which uses some form of chemical defense, the red imported fire ant triggered seven times more application of formic acid.

The authors hypothesize that formic acid rinses are an adaptive trait in crazy ants, evolved in its native range and employed when the two old foes are reunited. But formic acid rinses don't seal the crazy ant's advantage. Forty years ago, Bhatkar *et al.* (9) found that the lowly lawn ant *Lasius neoniger* (which also produces formic acid) can groom away a dose of fire ant poison; it just loses a chemical war of attrition to the more popu-

lous colonies of the fire ant. Crazy ants can achieve worker densities that are 100 times as high as those of species in the invaded habitat (2). Its antidote gives it the edge.

Biological control efforts often build on the premise that successful invasive species have escaped the parasites and predators of their native ecosystem (10). LeBrun *et al.* make a strong case that the red imported fire ant owes its long ride in the American South to its escape from a competitor. The crazy ant may be the fourth in a sequence of ant species that have hit the American Gulf Coast in the past century, each replacing the preceding as common and pernicious (2).

Given their ubiquity and impact (10), invasive ant species are model ecological systems for studying the many factors that regulate populations. As successive invasions reconstruct the population interactions of a South American ant community in South Texas, a logical next step is to search for the crazy ant's Achilles heel. One fruitful avenue may lie in evolutionary games of rock-paper-

scissors (4), where round robins of toxins and antidotes make the competitor of your competitor your friend. A more basic puzzle in our homogenizing world is why some—or perhaps all—disruptive invasions eventually crash (7, 11).

References

1. E. G. LeBrun *et al.*, *Science* **343**, 1014 (2014).
2. E. G. LeBrun, J. Abbott, L. E. Gilbert, *Biol. Invasions* **15**, 2429 (2013).
3. urbanentomology.tamu.edu/ants/raspberry.html
4. B. Kerr, M. A. Riley, M. W. Feldman, B. J. M. Bohannan, *Nature* **418**, 171 (2002).
5. L. Chao, B. R. Levin, *Proc. Natl. Acad. Sci. U.S.A.* **78**, 6324 (1981).
6. G. H. Orians, D. H. Janzen, *Am. Nat.* **108**, 581 (1974).
7. W. R. Tschinkel, *The Fire Ants* (Harvard Univ. Press, Cambridge, 2006).
8. D. H. Feener Jr. *et al.*, *Ecology* **89**, 1824 (2008).
9. A. Bhatkar, W. Whitcomb, W. Buren, P. Callahan, T. Carlyle, *Environ. Entomol.* **1**, 274 (1972).
10. D. Simberloff, *Invasive Species: What Everyone Needs to Know* (Oxford Univ. Press, Oxford, 2013).
11. M. Cooling, S. Hartley, D. A. Sim, P. J. Lester, *Biol. Lett.* **8**, 430 (2012).

10.1126/science.1251272

MATERIALS SCIENCE

The Surface Mobility of Glasses

Fei Chen,¹ Chi-Hang Lam,² Ophelia K. C. Tsui^{1,3}

The diffusion of atoms and molecules on a crystal surface plays an important role in myriad applications including thin-film deposition, sintering, and heterogeneous catalysis (1, 2). Surface diffusion is frequently observed at temperatures appreciably below the crystal's melting point, implying a role for enhanced surface mobility in the process. However, understanding the dynamics of surface diffusion in glasses is a research area still in its infancy. On page 994 of this issue, Chai *et al.* (3) present an experimental technique that enables detailed quantification of the near-surface mobility of glasses.

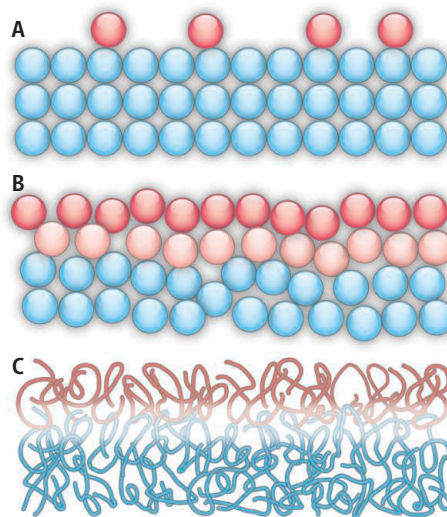
Although enhanced surface mobility was found by Chai *et al.* as well as by others in small-molecule and polymer glasses (4–7), there is a noteworthy distinction between these and the analogous observations in crystals. In crystals, the substrate surface is frequently much less mobile than the surface

Moving along. Mobile adatoms on a crystal surface (A) and their counterparts in the surface mobile layer of an organic glass (B) and a polymer glass (C). The mobile species are shown in red; the less mobile bulk-like species are in blue.

atoms or molecules (see the figure, panel A). In glasses, however, the first or several surface monolayers are molten even below the glass transition temperature T_g (where the glass freezes), and the change in dynamics from the surface is gradual (see the figure, panels B and C). The reason for such a difference may be that the temperatures commonly used in studies of glass surfaces are close to T_g . This proximity in temperature is attributable to a broad interest in connecting enhanced surface mobility, if present, with the anomalous T_g reduction observed in polymer films (8) and, more recently, fast organic crystal growth and the formation of ultrastable glasses (7).

Computer simulations have consistently revealed the presence of a surface mobile layer in glasses (9). Experimental verification has been made only recently. In one method, the relaxation time for the flattening of nano-dimples created on a polymer sur-

Surface diffusion on frozen polymer glasses is influenced by the surface dynamics of the glass itself.



face was measured (4). In another, polymer films were doped with fluorescent molecules whose dynamics are tied to those of the polymer (6); the relaxation time and relative population of the component exhibiting faster dynamics were measured. However, it is generally not straightforward to relate these relaxation times to familiar transport measures such as mobility or diffusivity. Typically, the mobility is determined by monitor-

¹Department of Physics, Boston University, Boston, MA 02215, USA. ²Department of Applied Physics, Hong Kong Polytechnic University, Hung Hom, Hong Kong. ³Division of Materials Science and Engineering, Boston University, Boston, MA 02215, USA. E-mail: oktsui@bu.edu; c.h.lam@polyu.edu.hk

ing the evolution of the surface morphology of a specimen and then analyzing the result by means of the Navier-Stokes equation (5, 10) or an equivalent model of fluid flow (7). The basic idea is that any nonflat surface structure (artificially or spontaneously created) produces pressure gradients that then drive the specimen to flow. In the lubrication approximation (usually applicable to thin-film specimens with thickness less than ~ 100 nm), the flow is planar and on average parallel to the pressure gradient. The current (or flow of fluid) per unit width is proportional to the pressure gradient and the film mobility, which can be used to determine the viscosity (5).

In one example study, the dynamics for the Brownian height fluctuations of an equilibrated film was monitored and modeled against that of overdamped surface capillary waves (10). In two others, surface structures, either shorter (5) or taller (7) than equilibrium, were created and the dissipative dynamics toward equilibrium (equivalent to that of the former example by the fluctuation-dissipation theorem) was monitored. To discern any anomalous surface mobility, the Navier-Stokes equation was solved for a bilayer film

comprising a mobile layer on top of a bulk-like layer. The solution predicts that a crossover from bulk flow to surface flow can occur by either decreasing the thickness or lowering the temperature. The former has been verified by systematically decreasing the thickness from 86 to 2 nm (5).

Chai *et al.* measured the flattening dynamics of a step edge created on the surface of polymer films with an average thickness around 100 nm. Upon cooling the films, they observed an analogous flow transition at T_g . A previous experiment (7) studying the flattening of surface gratings imprinted on micrometer-thick films of an organic glass also observed a transition from bulk diffusion [a mechanism only feasible in thick films (2)] to surface diffusion at $T_g + 12$ K upon cooling the films. All these findings reinforce the conclusion that surface diffusion is directly tied to the phenomenon of enhanced surface mobility of glasses. Indeed, it becomes the dominant transport process upon lowering the temperature or thinning the specimen.

It remains unknown whether surface diffusion is possible for long-chain polymers, particularly for those with radii of gyration

exceeding several nanometers [the thickness of the surface mobile region as derived from surface relaxation time studies (4, 6), which can reveal local motions besides surface flow]. Efforts to understand the dynamics of these materials will have to incorporate material viscoelasticity in the data analysis, which has hitherto been treated sparingly (11–13).

References and Notes

1. R. Gomer, *Rep. Prog. Phys.* **53**, 917 (1990).
2. W. W. Mullins, *J. Appl. Phys.* **30**, 77 (1959).
3. Y. Chai *et al.*, *Science* **343**, 994 (2014).
4. Z. Fakhraai, J. A. Forrest, *Science* **319**, 600 (2008).
5. Z. Yang *et al.*, *Science* **328**, 1676 (2010).
6. K. Paeng *et al.*, *J. Am. Chem. Soc.* **133**, 8444 (2011).
7. L. Zhu *et al.*, *Phys. Rev. Lett.* **106**, 256103 (2011).
8. J. L. Keddie, R. A. L. Jones, R. A. Cory, *Europhys. Lett.* **27**, 59 (1994).
9. J. Baschnagel, F. Varnik, *J. Phys. Condens. Matter* **17**, R851 (2005).
10. H. Kim *et al.*, *Phys. Rev. Lett.* **90**, 068302 (2003).
11. S. A. Hutcheson, G. B. McKenna, *Phys. Rev. Lett.* **94**, 076103 (2005).
12. M. Hamdorf, D. Johannsmann, *J. Chem. Phys.* **112**, 4262 (2000).
13. C.-H. Lam, O. K. C. Tsui, D. Peng, *Langmuir* **28**, 10217 (2012).

Acknowledgments: O.K.C.T. is supported by NSF through projects DMR-1004648 and DMR-1310536.

10.1126/science.1248113

CLIMATE CHANGE

The Tropical Pacific Ocean—Back in the Driver's Seat?

Amy Clement¹ and Pedro DiNezio²

Average temperatures at Earth's surface are now higher than they were in the mid-19th century, but the rate of warming has not been steady. A pause in surface warming in the mid-20th century coincided with increases in the atmospheric concentrations of sulfate aerosols, which are generally understood to cool the planet. Surface warming resumed in the 1970s, when strong pollution controls were implemented in developed countries. Thus, a balance of warming by greenhouse gases and cooling by aerosols may explain the variable rates of surface warming in the past century. A pause in global warming since 2000—a global warming “hiatus”—has opened up new questions about natural and human

activity-driven (anthropogenic) effects on global mean trends in surface temperature. Recent studies point to the importance of the tropical Pacific in driving these changes.

A range of factors may have contributed to the current pause in global warming, including changes in stratospheric water vapor, aerosol concentrations (1), and reductions in the Sun's output (2). The quantitative influence of these factors is still uncertain. However, what is striking about the current hiatus is that while many regions of the globe have continued to warm, the tropical Pacific has been colder than it was during the latter part of the 20th century.

In a recent study, Kosaka and Xie (3) showed that by prescribing the cold temperatures in this region (which represents



Challenges in
**CLIMATE
SCIENCE**
scim.ag/climatechall

Persistent cool conditions in the eastern tropical Pacific may explain the current global warming “hiatus.”

less than 10% of Earth's surface), their model can simulate the pause in global mean temperature since 2000, even when greenhouse gases have been increasing. In another climate model study, Meehl *et al.* found that a cold tropical Pacific increases the heat stored below the ocean surface, thus partially offsetting the warming at the surface (4). In the latter model, such hiatus periods arise as a result of natural variations in the climate system, implying that future global surface temperatures will be marked by periods of slowed and accelerated warming as a result of naturally occurring cold and warm periods in the tropical Pacific. Together, the two studies (3, 4) make a compelling case for a modulating effect of the Pacific.

Will these results hold up in other models? The answer depends on the Pacific's natural

¹Rosenstiel School of Marine and Atmospheric Science, University of Miami, 4600 Rickenbacker Causeway, Miami, FL 33149, USA. ²School of Ocean and Earth Science and Technology, University of Hawaii at Manoa, Honolulu, HI 96822, USA. E-mail: aclement@rsmas.miami.edu